

AI RECRUITING PLATFORM & ATS CORE

SOFTWARE REQUIREMENT SPECIFICATION (SRS) & ARCHITECTURE BLUEPRINT

Version: 2.4.0-PROD	Date: October 2026	Target Agent: Antigravity AI Engineer	Classification: Production Ready
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1. SYSTEM ARCHITECTURE & TOPOLOGY

He thong duoc kien truc theo mo hinh phan lop hien dai (Clean Multi-tier Architecture) tach biet ro rang giua cong khai va noi bo:

Phan He	Pham Vi & Cac Tinh Nang Chinh	Doi Tuong Su Dung
Employer Portal (Cong Nha Tuyen Dung)	- Dashboard KPI & Tong quan phau tuyen dung theo thoi gian thuc. - Quan ly Tin tuyen dung (Job Postings) & Thiet lap trong so AI. - Pipeline Kanban truc quan quan ly 7 giai doan ung vien. - Quan ly lich phong van, phong hop video truc tuyen. - Scorecard & Rubric Analyzer danh gia sau phong van. - Kho Nhan Tai (Talent Pool) & Semantic Rediscovery.	HR Manager, Recruiter, Interviewer, Hiring Board
Candidate Portal (Cong Tuyen Dung Cong Khai)	- Danh sach vi tri viec lam mo tuyen (/careers, /jobs/public). - Quick Apply khong bat buoc tao tai khoan, AI CV parsing tu dong. - Tra cuu trang thai ho so ung tuyen minh bach (/careers/track). - Chatbot AI Career Copilot tu van 24/7.	Ung vien dai chung, Nguoi tim viec
Recruitment Core / ATS	- Quan tri thuc the ung vien toan dien (Candidate 360). - Pipeline Stages (new -> reviewing -> interview -> offer -> hired). - Luu tru tai lieu CV, trich xuất vector embedding, nhat ky kiem toan.	Backend Engine & Database Layer
AI Layer (Gemini)	- CV Parser: Trich xuất hoc van, ky nang, kinh nghiem sang JSON. - Candidate-Job Matching: So khop ngu nghia, giai thich Strong/Gap. - Interview Question Generator: Goi y cau hoi phong van STAR. - Transcript Rubric Analyzer: Danh gia ghi am theo tieu chi. - Talent Rediscovery: Tim kiem va tai su dung ung vien tu Talent Pool.	Google Gemini API & LLM Orchestrator

2. CORE PRINCIPLE: CANDIDATE PROFILE INDEPENDENT OF APPLICATION

Nguyen tac bat bien cua kien truc ATS hien dai: Candidate Profile la mot thuc the doc lap duy nhat luu tru thong tin con nguoi (Ho ten, email, sdt, hoc van, ky nang, kinh nghiem, file goc). Moi lan nop ho so cho mot Job cu the se tao ra mot thuc the **Job Application** lien ket voi Candidate do.

- **Chong trung lap (Deduplication):** Khi ung vien da co trong he thong nop vao Job khac, he thong tu dong nhan dien email de tai su dung Profile, tao ban ghi Application moi, khong lam roi loan du lieu.
- **Kich hoat AI Talent Rediscovery:** Khi doanh nghiep mo Job moi, HR co the dung bo tim kiem ngu nghia de quet toan bo Kho Nhan Tai va tai ket noi (Rediscover) ung vien phu hop vao Job moi ma khong can dang bai tuyen dung lai tu dau.

3. SEPARATION OF CONCERNS (PHAN TACH TRACH NHIEM)

Phan He	Trach Nhiem Chuyen Biet	Ranh Gioi Kien Truc
Workflow Engine	Dieu phoi trang thai pipeline (Applied -> Reviewing -> Interview -> Offer -> Hired/Rejected). Dong bo lich hen va thong bao.	FastAPI State Machine & Celery Workers
Rule Engine	Kiem tra quy tac dinh luong cung: muc luong vuot tran ngan sach, bat buoc hoan thanh phong van va co scorecard truoc khi phat hanh thu moi offer.	Deterministic Business Logic (Python Services)
AI Copilot	Doc hieu ngu nghia ngon ngu tu nhien, trich xuat CV, cham diem doi sanh JD, phan tich noi dung boc bang phong van. Chi dong vai tro co van, khong ra quyet dinh thay con nguoi.	Gemini LLM Client with Schema Validation

4. ETHICAL AI GUARDRAILS (RAO CHAN DAO DUC AI)

QUY TAC DAO DUC BAT DI BAT DICH CUA NEN TANG:

1. **KHONG** su dung AI nhan dien khuon mat hoac cham diem cam xuc/thai do cua ung vien.
2. **KHONG** phan tich am sac giong noi de suy dien tinh cach hay tam ly.
3. **KHONG** tu dong loai ho so (Auto-reject) ma khong co su xem xet va phe duyet truc tiep cua chuyen vien HR (Human-in-the-loop).
4. **KHONG** dua ra diem so mu mo: Moi diem tuong thich deu phai di kem giai thich cu the cac diem khop manh (Strong Matches) va cac diem con thieu hut (Missing Evidence).

5. ANTIGRAVITY ENGINEERING TASK CHECKLIST (100% COMPLETE)

Trang Thai	Hang Muc Tinh Nang Theo Dac Ta SRS	Vi Tri Codebase / Endpoint
[x] HOAN TAT	Dual Portals: Tach biet tuyet doi Cong Nha Tuyen Dung va Cong Tuyen Dung Cong Khai	components/layout/AppHeader.tsx, app/careers
[x] HOAN TAT	Quick Apply & AI CV Parser: Nop ho so khong can dang ky tai khoan, tu dong boc tach CV	POST /api/v1/candidates/apply, app/(public)/apply
[x] HOAN TAT	Candidate Status Tracking: Tra cuu trang thai ho so truc tuyen theo email minh bach	GET /api/v1/candidates/track/status, /careers/track
[x] HOAN TAT	24/7 AI Career Chatbot: Tro ly ao ho tro ung vien tim viec va giai dap quy trinh	POST /api/v1/candidates/career-chat, CareerChatWidget.tsx

[x] HOAN TAT	Configurable Kanban Pipeline: Dieu phoi 7 trang thai ung vien truc quan	PATCH /candidates/applications/.../pipeline-status
[x] HOAN TAT	Interview & Transcript Rubric: Quan ly phong hop truc tuyen va danh gia transcript	app/(dashboard)/interviews, /evaluations
[x] HOAN TAT	Talent Pool & AI Rediscovery: Tim kiem ngu nghia va tai ket noi ung vien vao Job moi	POST /candidates/{id}/rediscover, /talent-pool
[x] HOAN TAT	Export SRS PDF: Xuat tai lieu ban dac ta ky thuat truc tiep phuc vu ban giao	scripts/generate_srs_pdf.py, public/AI_RECRUITING_SRS...

XAC NHAN BAN GIAO KY THUAT (ENGINEERING HANDOFF SIGN-OFF):
Tai lieu va toan bo ma nguon he thong da duoc kiem thu chat che voi Neon PostgreSQL, Next.js 15, FastAPI va Google Gemini AI.
San pham san sang trien khai van hanh Production.